ETHENA · SECURITY REVIEW

Ethena EthenaLock.

A security review of EthenaLock.

CLIENT
Ethena

PROJECT
EthenaLock

AUDITORS
Jakub Zmyslowski,
Vlad Zotov,
Michael Lett,
Ferran Celades

AUDIT WINDOW
September 30 –
October 6, 2026

FINAL REPORT
October 6, 2026

§ 00

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12 pages: overview, confidence ranking, scope, 3 findings, disclaimer, and about Guardian.

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§ 01

Overview.

Guardian reviewed EthenaLock across 2 review rounds. The final report includes 3 Informational findings.

AUDIT FIRM
Guardian

CLIENT
Ethena

PROJECT
EthenaLock

PREPARED BY
Jakub Zmyslowski, Vlad Zotov, Michael Lett, Ferran Celades

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REPOSITORY
`https://github.com/GuardianOrg/mirror-chainnodesorg-ethenapay-wallet-contracts-4705b4da`

METHODOLOGY
Smart contract · Manual · Main Review · Remediation Review

MAIN REVIEW COMMIT
`64a8bf610090`

REMEDIATION REVIEW COMMIT
`e1a12efe0ae8`

1.1 Confidence Ranking

5 / 5

Very High Confidence

Guardian assigns a Confidence Ranking of 5 out of 5, combining finding severity, remediation outcome, and the reviewer's assessment of latent risk. The full scale is on the next page.

1	2	3	4	5
VERY LOW CONFIDENCE	LOW CONFIDENCE	MODERATE CONFIDENCE	HIGH CONFIDENCE	VERY HIGH CONFIDENCE

1.2 Findings at a Glance

SEVERITY	TOTAL	PENDING	ACK	PARTIAL	RESOLVED
Critical	**0**	0	0	0	0
High	**0**	0	0	0	0
Medium	**0**	0	0	0	0
Low	**0**	0	0	0	0
Informational	**3**	0	0	0	3
All severities	**3**	**0**	**0**	**0**	**3**

§ 02

Confidence Ranking.

Guardian assigns a project-level confidence score using the scale below. The ranking combines finding severity, remediation outcome, and the reviewer's assessment of latent risk.

CONFIDENCE RANKING	DEFINITION AND RECOMMENDATION	RISK PROFILE
5 Very High Confidence ASSIGNED	Codebase is mature, clean, and secure. No High or Critical vulnerabilities were found. Follows modern best practices with high test coverage and thoughtful design. Recommendation: code is highly secure at time of audit.	0 High/Critical findings and few Low/Medium severity findings.
4 High Confidence	Code is clean, well-structured, and adheres to best practices. Design patterns are sound and test coverage is strong. Recommendation: suitable for deployment after remediations; consider periodic review with changes.	0-1 High/Critical findings per engagement week and little to no Medium severity findings.
3 Moderate Confidence	Code is functional but shows gaps in defensive programming or test coverage. Recommendation: remediate the findings and re-review the changes before deployment.	Some High/Critical findings, with various Medium severity findings.
2 Low Confidence	Recurring issues suggest systemic gaps rather than isolated mistakes. Recommendation: remediate, then commission a follow-up review at a frozen commit.	Over 6 High and Critical findings across the engagement.
1 Very Low Confidence	The volume and severity of findings indicate the codebase is not ready for production. Recommendation: substantial rework, then a full re-review.	Over 12 High and Critical findings across the engagement.

§ 03

Scope & Classifications.

Guardian reviewed EthenaLock in `https://github.com/GuardianOrg/mirror-chainnodesorg-ethenapay-wallet-contracts-4705b4da`. The scope includes 230 source lines across 2 files.

3.1 Severity Matrix

	IMP · HIGH	IMP · MED	IMP · LOW
LIK · HIGH	Critical	High	Medium
LIK · MED	High	Medium	Low
LIK · LOW	Medium	Low	Low

IMPACT · HIGH	Significant loss of assets, harm to a group of users, or core protocol functionality is disrupted.
IMPACT · MEDIUM	Small amount of funds can be lost, or ancillary functionality is affected. Users may experience reduced or delayed receipt.
IMPACT · LOW	Unexpected behavior in some functionality that is notable but does not meet the criteria for higher severity.
LIKELIHOOD · HIGH	Possible with reasonable assumptions that mimic on-chain conditions; cost is low compared to amount gained.
LIKELIHOOD · MEDIUM	Possible only in uncommon cases or requires a large amount of capital relative to disruption caused.
LIKELIHOOD · LOW	Unlikely to ever occur in production.

Files in Scope.

FILE	NSLOC	TOTAL
script/DeployEthenaLock.s.sol	60	92
src/EthenaLock.sol	170	294

§ 04

Findings.

Grouped by round. Visible IDs reset by severity within each round, so a finding is named the same way in the report as it was in the channel it was written in. Status is where each finding ended up; its own page says how.

ID	TITLE	CATEGORY	SEVERITY	STATUS	PAGE
MAIN REVIEW					
I-01	Incorrect NatSpec for ZeroAddress	Documentation	INFORMATIONAL	RESOLVED	p. 8
I-02	Incorrect permanent insolvency claim	Documentation	INFORMATIONAL	RESOLVED	p. 9
I-03	Misleading custom error argument names	Documentation	INFORMATIONAL	RESOLVED	p. 10

INFORMATIONAL I-01 RESOLVED

Incorrect NatSpec for ZeroAddress

CATEGORY	Documentation
LOCATION	`src/EthenaLock.sol:21-22`
REVIEW	Main Review

DESCRIPTION

The NatSpec for `ZeroAddress` incorrectly states that it is raised when the constructor receives a zero `admin_` address. That condition actually raises the inherited `OwnableInvalidOwner` error.

`ZeroAddress` is used when the ENA token address, the wallet passed to `sunsetRelease`, or the recipient passed to `rescue` is zero.

This mismatch can mislead integrators relying on generated documentation to identify and handle reverts.

RECOMMENDATION

Update the NatSpec to describe the actual conditions that raise `ZeroAddress`.

RESOLUTION

Status: Resolved.

Ethena Team: Fixed in 757f2d1

INFORMATIONAL I-02

RESOLVED

Incorrect permanent insolvency claim

CATEGORY	Documentation
LOCATION	docs/ETHENA_LOCK.md:191-203, 429-435
REVIEW	Main Review

DESCRIPTION

The EthenaLock documentation states:

> A shortfall of even 1 wei freezes every position permanently, with no owner lever

However, `_assertSolvent` compares the current token balance against `totalLockedAll`. Anyone can transfer tokens directly to the contract without increasing `totalLockedAll` or creating a position. Once the transfer covers the shortfall, matured withdrawals can succeed again, assuming normal token transfers.

This description could mislead operators during incident response, causing them to overlook a valid recovery path and treat recoverable funds as permanently inaccessible.

RECOMMENDATION

Update the documentation to explain that withdrawals remain blocked while the contract is insolvent and that direct token transfers sufficient to cover the deficit can restore solvency.

RESOLUTION

Status: Resolved.

Ethena Team: Fixed in f30e52e

INFORMATIONAL I-03 RESOLVED

Misleading custom error argument names

CATEGORY	Documentation
LOCATION	`src/EthenaLock.sol:23-27`
REVIEW	Main Review

DESCRIPTION

`BelowMinLock` and `AmountOverflow` name their first argument `received`, but the reported value does not always represent tokens received during a deposit.

`BelowMinLock` can report the requested deposit amount, the actual amount received, or the remaining locked balance after a withdrawal. `AmountOverflow` reports the total position balance after adding the deposit.

These misleading names may cause integrators to misinterpret decoded revert data.

RECOMMENDATION

Consider renaming the argument to `amount` in `BelowMinLock` and `positionAmount` in `AmountOverflow`.

RESOLUTION

Status: Resolved.

Ethena Team: Fixed in e1a12ef

§ 05

Disclaimer.

This report is not, nor should be considered, an "endorsement" or "disapproval" of any particular project or team. This report is not, nor should be considered, an indication of the economics or value of any "product" or "asset" created by any team or project that contracts Guardian to perform a security assessment. This report does not provide any warranty or guarantee regarding the absolute bug-free nature of the technology analyzed, nor do they provide any indication of the technologies proprietors, business, business model or legal compliance.

This report should not be used in any way to make decisions around investment or involvement with any particular project. This report in no way provides investment advice, nor should be leveraged as investment advice of any sort. This report represents an extensive assessing process intending to help our customers increase the quality of their code while reducing the high level of risk presented by cryptographic tokens and blockchain technology.

Blockchain technology and cryptographic assets present a high level of ongoing risk. Guardian's position is that each company and individual are responsible for their own due diligence and continuous security. Guardian's goal is to help reduce the attack vectors and the high level of variance associated with utilizing new and consistently changing technologies, and in no way claims any guarantee of security or functionality of the technology we agree to analyze.

The assessment services provided by Guardian is subject to dependencies and under continuing development. You agree that your access and/or use, including but not limited to any services, reports, and materials, will be at your sole risk on an as-is, where-is, and as-available basis.

Cryptographic tokens are emergent technologies and carry with them high levels of technical risk and uncertainty. The assessment reports could include false positives, false negatives, and other unpredictable results. The services may access, and depend upon, multiple layers of third-parties.

Notice that smart contracts deployed on the blockchain are not resistant from internal/external exploit. Notice that active smart contract owner privileges constitute an elevated impact to any smart contract's safety and security. Therefore, Guardian does not guarantee the explicit security of the audited smart contract, regardless of the verdict.

§ 06

About Guardian.

Guardian is a smart contract security firm. Every engagement is worked by two independent teams competing on the same scope, each with its own tooling and methods, so no single reviewer's blind spot survives.

$45B+
DIGITAL ASSETS UNDER REVIEW

200+
ENGAGEMENTS COMPLETED

500+
CRITICAL & HIGH FINDINGS REPORTED

Reach us at audits@guardianaudits.com.